

Horner's

To evaluate the polynomial

$$P(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0 = (x - x_0)Q(x) + b_0$$

and its derivative at x_0 :

INPUT degree n ; coefficients $a_0, a_1, \dots, a_n; x_0$.

OUTPUT $y = P(x_0); z = P'(x_0)$.

Step 1 Set $y = a_n$; (Compute b_n for P .)
 $z = a_n$. (Compute b_{n-1} for Q .)

Step 2 For $j = n - 1, n - 2, \dots, 1$
 set $y = x_0 y + a_j$; (Compute b_j for P .)
 $z = x_0 z + y$. (Compute b_{j-1} for Q .)

Step 3 Set $y = x_0 y + a_0$. (Compute b_0 for P .)

Step 4 OUTPUT (y, z) ;
 STOP. ■

If the N th iterate, x_N , in Newton's method is an approximate zero for P , then

$$P(x) = (x - x_N)Q(x) + b_0 = (x - x_N)Q(x) + P(x_N) \approx (x - x_N)Q(x),$$

so $x - x_N$ is an approximate factor of $P(x)$. Letting $\hat{x}_1 = x_N$ be the approximate zero of P and $Q_1(x) \equiv Q(x)$ be the approximate factor gives

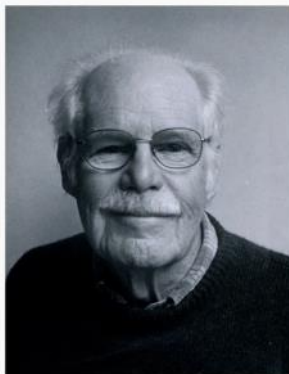
$$P(x) \approx (x - \hat{x}_1)Q_1(x).$$

We can find a second approximate zero of P by applying Newton's method to $Q_1(x)$.

$$P(x) \approx (x - \hat{x}_1)(x - \hat{x}_2) \cdots (x - \hat{x}_k)Q_k(x).$$

Complex Zeros: Müller's Method

One problem with applying the Secant, False Position, or Newton's method to polynomials is the possibility of the polynomial having complex roots even when all the coefficients are real numbers. If the initial approximation is a real number, all subsequent approximations will also be real numbers. One way to overcome this difficulty is to begin with a complex initial approximation and do all the computations using complex arithmetic. An alternative approach has its basis in the following theorem.



David E. Muller
1924 - 2008

Photo & Bio Courtesy of NMSU

- PhD Physics - California Institute of Technology 1951
- University of Illinois 1953 - 1992
 - Professor of Mathematics and Computer Science
 - Supervised 17 doctoral theses
- New Mexico State University (NMSU) 1995 - 2008
 - Adjunct Professor of Mathematics

“A Method for Solving Algebraic Equations Using an Automatic Computer” By David E. Muller

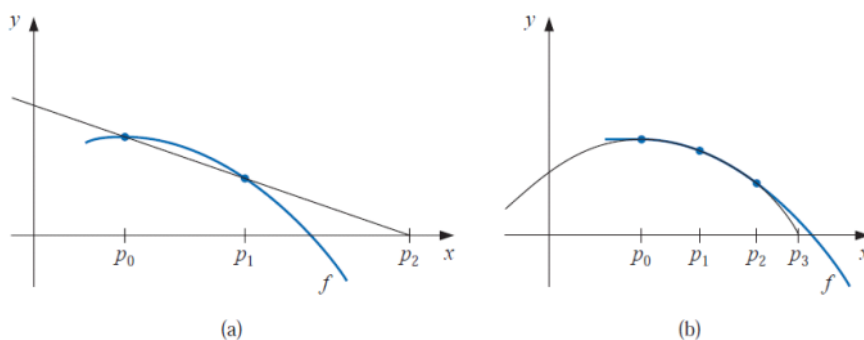
- *Mathematical Tables and Other Aids to Computation*, Vol. 10, No. 56 (Oct., 1956), pp. 208-215. American Mathematical Society.
- Cited by 700+
- “Automatic Computer” ILLIAC I
- 5 tons – 10ft L × 2ft W × 8½ft H (3m×.6m×2.6m)
- 2,800 vacuum tubes
- 1,024 40-bit words Memory & 12,800 40-bit words Drum Storage
- Paper measures computation time in minutes

Secant Method but with a Parabola

- Find root of continuous function $f(x)$
- Three starting values x_1, x_2 , and x_3
- Compute curve between $(x_1, f(x_1))$, $(x_2, f(x_2))$, and $(x_3, f(x_3))$
- Test both x intercepts
- Keep the one where function is closer to zero at x_4
- Repeat using newest three x 's until you find root $|f(x_n)| < \epsilon$

Theorem 2.20 If $z = a + bi$ is a complex zero of multiplicity m of the polynomial $P(x)$ with real coefficients, then $\bar{z} = a - bi$ is also a zero of multiplicity m of the polynomial $P(x)$, and $(x^2 - 2ax + a^2 + b^2)^m$ is a factor of $P(x)$. ■

Müller's method is similar to the Secant method. But whereas the Secant method uses a line through two points on the curve to approximate the root, Müller's method uses a parabola through three points on the curve for the approximation.



The derivation of Müller's method begins by considering the quadratic polynomial

$$P(x) = a(x - p_2)^2 + b(x - p_2) + c$$

that passes through $(p_0, f(p_0))$, $(p_1, f(p_1))$, and $(p_2, f(p_2))$. The constants a , b , and c can be determined from the conditions

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$$f(p_0) = a(p_0 - p_2)^2 + b(p_0 - p_2) + c, \quad (2.17)$$

$$f(p_1) = a(p_1 - p_2)^2 + b(p_1 - p_2) + c, \quad (2.18)$$

and

$$f(p_2) = a \cdot 0^2 + b \cdot 0 + c = c \quad (2.19)$$

to be

$$c = f(p_2), \quad (2.20)$$

$$b = \frac{(p_0 - p_2)^2[f(p_1) - f(p_2)] - (p_1 - p_2)^2[f(p_0) - f(p_2)]}{(p_0 - p_2)(p_1 - p_2)(p_0 - p_1)}, \quad (2.21)$$

and

$$a = \frac{(p_1 - p_2)[f(p_0) - f(p_2)] - (p_0 - p_2)[f(p_1) - f(p_2)]}{(p_0 - p_2)(p_1 - p_2)(p_0 - p_1)}. \quad (2.22)$$

To determine p_3 , a zero of P , we apply the quadratic formula to $P(x) = 0$. However, because of round-off error problems caused by the subtraction of nearly equal numbers, we apply the formula in the manner prescribed in Eq (1.2) and (1.3) of Section 1.2:

$$p_3 - p_2 = \frac{-2c}{b \pm \sqrt{b^2 - 4ac}}$$

This formula gives two possibilities for p_3 , depending on the sign preceding the radical term. In Müller's method, the sign is chosen to agree with the sign of b . Chosen in this manner, the denominator will be the largest in magnitude and will result in p_3 being selected as the closest zero of P to p_2 . Thus

$$p_3 = p_2 - \frac{2c}{b + \operatorname{sgn}(b)\sqrt{b^2 - 4ac}} \quad p_3 = 0 - \frac{2(1)}{0.25 + \sqrt{(0.25)^2 - 4(1.25)(1)}}$$

where a , b , and c are given in Eqs. (2.20) through (2.22).

Easy Formula

$$h_0 = p_1 - p_0, \quad h_1 = p_2 - p_1,$$

$$\delta_0 = \frac{f(p_1) - f(p_0)}{h_0},$$

$$\delta_1 = \frac{f(p_2) - f(p_1)}{h_1}.$$

$$a = \frac{\delta_1 - \delta_0}{h_1 + h_0}$$

$$b = ah_1 + \delta_1$$

$$c = f(p_2).$$

$p_0 = -0.5, p_1 = 0.5, p_2 = 0$

$h_0 = -1, h_1 = 0.5$

$\delta_0 = \frac{f(-0.5) - f(0.5)}{-1}$

$\delta_1 = \frac{f(0) - f(-0.5)}{0.5}$

$a = 1.25$

$b = 0.25$

$c = 1$

Müller's

To find a solution to $f(x) = 0$ given three approximations, p_0 , p_1 , and p_2 :

INPUT p_0, p_1, p_2 ; tolerance TOL ; maximum number of iterations N_0 .

OUTPUT approximate solution p or message of failure.

Step 1 Set $h_1 = p_1 - p_0$;
 $h_2 = p_2 - p_1$;
 $\delta_1 = (f(p_1) - f(p_0))/h_1$;
 $\delta_2 = (f(p_2) - f(p_1))/h_2$;
 $d = (\delta_2 - \delta_1)/(h_2 + h_1)$;
 $i = 3$.

Step 2 While $i \leq N_0$ do Steps 3–7.

Step 3 $b = \delta_2 + h_2 d$;
 $D = (b^2 - 4f(p_2)d)^{1/2}$. (Note: May require complex arithmetic.)

Step 4 If $|b - D| < |b + D|$ then set $E = b + D$
 else set $E = b - D$.

Step 5 Set $h = -2f(p_2)/E$;
 $p = p_2 + h$.

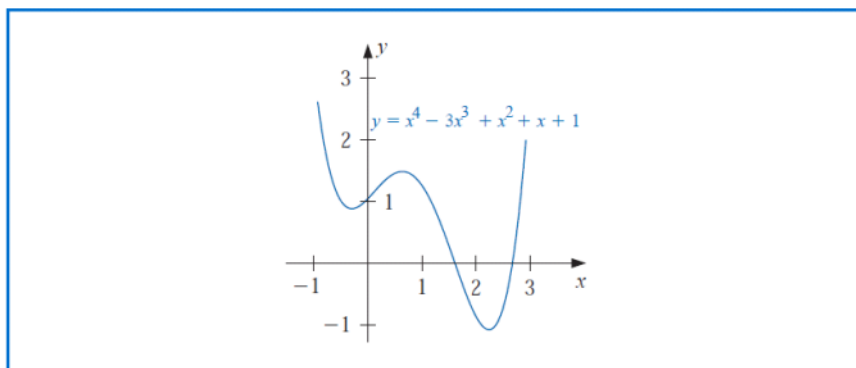
Step 6 If $|h| < TOL$ then
 OUTPUT (p); (The procedure was successful.)
 STOP.

Step 7 Set $p_0 = p_1$; (Prepare for next iteration.)
 $p_1 = p_2$;
 $p_2 = p$;
 $h_1 = p_1 - p_0$;
 $h_2 = p_2 - p_1$;
 $\delta_1 = (f(p_1) - f(p_0))/h_1$;
 $\delta_2 = (f(p_2) - f(p_1))/h_2$;
 $d = (\delta_2 - \delta_1)/(h_2 + h_1)$;
 $i = i + 1$.

Step 8 OUTPUT ('Method failed after N_0 iterations, $N_0 =$, N_0);
 (The procedure was unsuccessful.)
 STOP.

Illustration Consider the polynomial $f(x) = x^4 - 3x^3 + x^2 + x + 1$, part of whose graph is shown in Figure 2.14.

Figure 2.14



Three sets of three initial points will be used with Algorithm 2.8 and $TOL = 10^{-5}$ to approximate the zeros of f . The first set will use $p_0 = 0.5$, $p_1 = -0.5$, and $p_2 = 0$. The parabola passing through these points has complex roots because it does not intersect the x -axis. Table 2.12 gives approximations to the corresponding complex zeros of f .

Table 2.12

$p_0 = 0.5, p_1 = -0.5, p_2 = 0$		
i	p_i	$f(p_i)$
3	$-0.100000 + 0.888819i$	$-0.01120000 + 3.014875548i$
4	$-0.492146 + 0.447031i$	$-0.1691201 - 0.7367331502i$
5	$-0.352226 + 0.484132i$	$-0.1786004 + 0.0181872213i$
6	$-0.340229 + 0.443036i$	$0.01197670 - 0.0105562185i$
7	$-0.339095 + 0.446656i$	$-0.0010550 + 0.000387261i$
8	$-0.339093 + 0.446630i$	$0.000000 + 0.000000i$
9	$-0.339093 + 0.446630i$	$0.000000 + 0.000000i$

Table 2.13 gives the approximations to the two real zeros of f . The smallest of these uses $p_0 = 0.5, p_1 = 1.0$, and $p_2 = 1.5$, and the largest root is approximated when $p_0 = 1.5, p_1 = 2.0$, and $p_2 = 2.5$.

Table 2.13

$p_0 = 0.5, p_1 = 1.0, p_2 = 1.5$			$p_0 = 1.5, p_1 = 2.0, p_2 = 2.5$		
i	p_i	$f(p_i)$	i	p_i	$f(p_i)$
3	1.40637	-0.04851	3	2.24733	-0.24507
4	1.38878	0.00174	4	2.28652	-0.01446
5	1.38939	0.00000	5	2.28878	-0.00012
6	1.38939	0.00000	6	2.28880	0.00000
			7	2.28879	0.00000

We used Maple to generate the results in Table 2.12. To find the first result in the table, define $f(x)$ with

$$f := x \rightarrow x^4 - 3x^3 + x^2 + x + 1$$

Then enter the initial approximations with

$$p0 := 0.5; p1 := -0.5; p2 := 0.0$$

and evaluate the function at these points with

$$f0 := f(p0); f1 := f(p1); f2 := f(p2)$$

To determine the coefficients a, b, c , and the approximate solution, enter

$$c := f2;$$

$$b := \frac{((p0 - p2)^2 \cdot (f1 - f2) - (p1 - p2)^2 \cdot (f0 - f2))}{(p0 - p2) \cdot (p1 - p2) \cdot (p0 - p1)}$$

$$a := \frac{((p1 - p2) \cdot (f0 - f2) - (p0 - p2) \cdot (f1 - f2))}{(p0 - p2) \cdot (p1 - p2) \cdot (p0 - p1)}$$

$$p3 := p2 - \frac{2c}{b + \left(\frac{b}{\text{abs}(b)}\right)\sqrt{b^2 - 4a \cdot c}}$$

This produces the final Maple output

$$-0.1000000000 + 0.8888194418i$$

and evaluating at this approximation gives $f(p3)$ as

$$-0.0112000001 + 3.014875548i$$

- Faster than Secant Method
- Slower than Newton's Method
- Does not require a derivative
- Good at finding complex roots

Muller's Method Example

$$f(x) = x^3 - x^2 - x - 1$$

$$\epsilon = 10^{-7}$$

$$1) x_1 = 0, x_2 = 1, x_3 = 2$$

$$2) x_1 = -2, x_2 = -1, x_3 = 0$$

$$\delta_0 = \frac{f(p_1) - f(p_0)}{h_0}.$$

$$\delta_0 = \frac{1.1875 - 1.4375}{-1} = \boxed{0.25}.$$

$$\begin{aligned}\delta_1 &= \frac{f(p_2) - f(p_1)}{h_1} \\ &= \frac{1 - 1.1875}{0.5} = \boxed{-0.375}.\end{aligned}$$

$$a = \frac{\delta_1 - \delta_0}{h_1 + h_0}.$$

$$\begin{aligned}a &= \frac{-0.375 - 0.25}{0.5 - 1} \\ &= \frac{-0.625}{-0.5} = \boxed{1.25}.\end{aligned}$$

$$b = ah_1 + \delta_1.$$

$$\begin{aligned}b &= (1.25)(0.5) - 0.375 \\ &= 0.625 - 0.375 = \boxed{0.25}.\end{aligned}$$

$$\boxed{a = 1.25, \quad b = 0.25, \quad c = 1}.$$

$$\begin{aligned}
 b^2 - 4ac &= (0.25)^2 - 4(1.25)(1) \\
 &= 0.0625 - 5 \\
 &= -4.9375.
 \end{aligned}$$

$$\sqrt{-4.9375} = i\sqrt{4.9375} \approx 2.2220486i.$$

$$p_3 \approx -0.1 - 0.888819i.$$

This is an important point.

The discriminant was

$$b^2 - 4ac = -4.9375 < 0.$$

Therefore, the quadratic interpolation has **complex roots**.

So Müller naturally moves into the complex plane.

This is one of the major advantages of Müller's method: **it can find complex roots without explicitly starting with complex initial guesses.**

$$p_1 = -0.5, \quad p_2 = 0, \quad p_3 = -0.1 - 0.888819i.$$

$$h_1 = p_2 - p_1,$$

$$h_1 = 0 - (-0.5) = 0.5.$$

$$h_2 = p_3 - p_2,$$

$$h_2 = -0.1 - 0.8888194417i.$$

$$h_2 = -0.1 - 0.8888194417i.$$

$$\delta_1 = \frac{f(p_2) - f(p_1)}{h_1}.$$

$$f(p_2) = f(0) = 1$$

$$f(p_1) = f(-0.5) = 1.1875.$$

$$\delta_1 = \frac{1 - 1.1875}{0.5}$$

$$\boxed{\delta_1 = -0.375}.$$

$$\delta_2 = \frac{f(p_3) - f(p_2)}{h_2}.$$

$$p_3 = -0.1 - 0.8888194417i,$$

$$f(x) = x^4 - 3x^3 + x^2 + x + 1$$

$$f(p_3) \approx -0.0112 - 3.0148755464i.$$

$$\delta_2 = \frac{-0.0112 - 3.0148755464i}{-0.1 - 0.8888194417i}.$$

$$\boxed{\delta_2 \approx 3.8369999999 - 0.9750661299i}.$$

$$a = \frac{\delta_2 - \delta_1}{h_2 + h_1}.$$

$$a = \frac{(3.837 - 0.97506613i) - (-0.375)}{(-0.1 - 0.88881944i) + 0.5}.$$

$$a = \frac{4.212 - 0.97506613i}{0.4 - 0.88881944i}.$$

$$\boxed{a \approx 2.32 + 3.2886319344i}.$$

$$b = ah_2 + \delta_2.$$

$$b \approx 6.167 - 3.1375326293i.$$

$$c = f(p_3).$$

$$c = -0.0112 - 3.0148755464i.$$

$$\begin{aligned} a &= 2.32 + 3.2886319344i, \\ b &= 6.167 - 3.1375326293i, \\ c &= -0.0112 - 3.0148755464i. \end{aligned}$$

$$p_4 = p_3 - \frac{2c}{b \pm \sqrt{b^2 - 4ac}}.$$

$$b^2 - 4ac.$$

$$\sqrt{b^2 - 4ac} \approx 1.4416827772 - 3.6668786768i.$$

$$b + \sqrt{b^2 - 4ac}$$

$$b - \sqrt{b^2 - 4ac}.$$

$$\begin{aligned} (6.167 - 3.1375326293i) + (1.4416827772 - 3.6668786768i) \\ = 7.6086827772 - 6.8044113061i. \end{aligned}$$

$$4.7253172228 + 0.5293460475i.$$

We choose the denominator with the **larger absolute value**:

$$7.6086827772 - 6.8044113061i.$$

$$p_4 = p_3 - \frac{2c}{7.6086827772 - 6.8044113061i}.$$

$$p_3 = -0.1 - 0.8888194417i$$

$$c = -0.0112 - 3.0148755464i,$$

$$\boxed{p_4 \approx -0.4921457099 - 0.4470307000i}.$$